Physics 468/668: Statistical Physics **Spring 2019**

Lecture 17: Correlation and Scatterings

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17.1 Spatial correlation function

For a system with N particles, we define the *one and two number density* as follows,

$$\begin{aligned} \rho^{(1)}(r) &= \sum_{i=1}^{N} \delta(r - r_i) \\ \rho^{(2)}(r, r') &= \sum_{i=1}^{N-1} \sum_{j=i+1}^{N} \delta(r - r_i)\delta(r' - r_j) \end{aligned} \tag{17.1}$$

Clearly, the integral over all r space is simple number of atoms, and the number of atomic pairs.

$$\begin{aligned} \int \rho^{(1)}(r) dr &= N \\ \int \rho^{(2)}(r, r') dr dr' &= N(N-1) \end{aligned} \tag{17.2}$$

Higher body terms like $\rho^{(3)}$ can be obtained as well. Now we define the general correlation function $g^{(n)}$ as

$$\begin{aligned} g^{(n)}(r_1, \cdots, r_n) &= \frac{1}{\rho^n} \rho^{(n)}(r_1, \cdots, r_n) \\ &= \frac{1}{Z_N} \left(\frac{V}{N}\right)^n \frac{N!}{(N-n)!} \int dr_{n+1} \cdots dr_N e^{-\beta U(r_1, \cdots, r_N)} \end{aligned} \tag{17.3}$$

For $n = 2$, this leads to a so called radial distribution function $g^2(r_1, r_2)$,

$$\begin{aligned} g^2(r_1, r_2) &= \frac{\rho^{(2)}(r_1, r_2)}{\rho(r_1)\rho(r_2)} \\ &= \frac{1}{Z_N} \frac{V^2(N-1)}{N} \int dr_3 \cdots dr_N e^{-\beta U(r_1, \cdots, r_N)} \end{aligned} \tag{17.4}$$

For an homogeneous isotropic system, $g(r_1, r_2)$ only depends on $r_1 - r_2$, so

$$g^2(r_1, r_2) = g(|r_1 - r_2|) = \frac{\rho_N^{(2)}(r_1 - r_2)}{\rho^2} \tag{17.5}$$

This above equation used to compute $g(r)$ in numerical simulations such as molecular dynamics. So there will be two versions in which r can be either vector or scalar form. For the vector form

$$g(\boldsymbol{r}) = \frac{1}{\rho^2}\sum_{i=1}^{N-1}\sum_{j=i+1}^{N}\delta(\boldsymbol{r}-\boldsymbol{r_i})\delta(\boldsymbol{r'}-\boldsymbol{r_j}) = \frac{1}{N\rho}\sum_{i=1}^{N-1}\sum_{j=i+1}^{N}\delta(\boldsymbol{r}-\boldsymbol{r_i}+\boldsymbol{r_j}) \tag{17.6}$$

A more widely used case is the scalor form,

$$g(r) = \frac{1}{4\pi r^2}\frac{1}{N\rho}\sum_{i=1}^{N-1}\sum_{j=i+1}^{N}\delta(r-|r_i-r_j|) \tag{17.7}$$

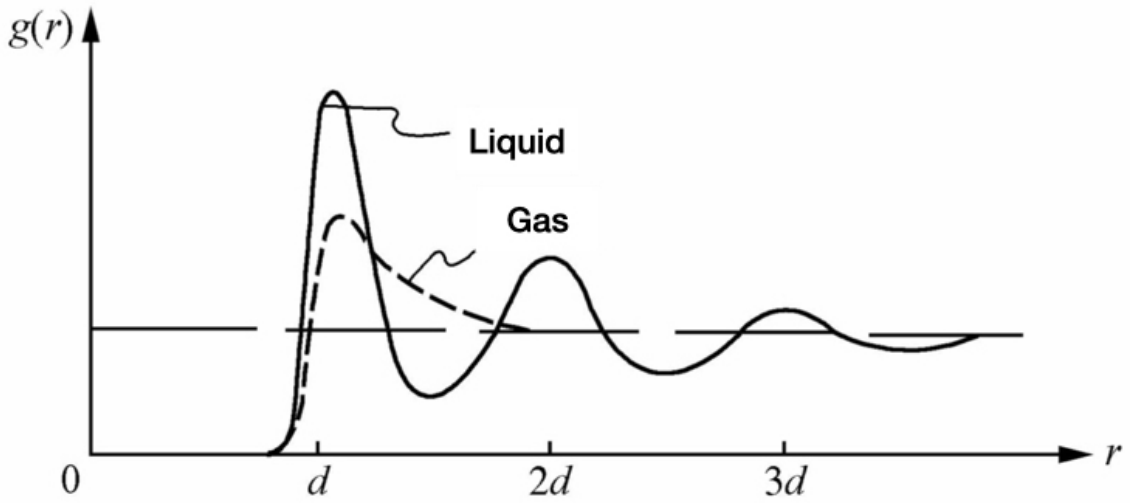
From the above equation, it is not difficult to obtain the relation that

$$\int_0^\infty 4\pi r^2 N\rho g(r)dr = N(N-1) \tag{17.8}$$

So,

$$\int_0^\infty 4\pi r^2 \rho g(r)dr = N-1 \tag{17.9}$$

Below is a typical plot of $g(r)$ for liquid and gas argon. Clearly, $g(r)$ is a spectrum with several peaks. It is always non-negative, also tends to converges to 1 in the large distances (which can be explained by its definition).



It tends out that $g(r)$ is a useful descriptor to tell the degree of order in terms of atomic packing in the 3D space. When it is ideal gas, $g(r)$ is always 1. When it is liquid, you will observe some peaks. When it is solid, the peaks will become sharper. If you integrate $g(r)$ from 0 to ∞, you get $N-1$. If you integrate from 0 to a given truncation, it tells you the number of particle within this truncation.

For dilute systems (e.g. gases), the correlations in the positions of the particles that $g(r)$ accounts for are only due to the potential $u(r)$ engendered by the reference particle, neglecting indirect effects. In the first approximation, it is thus simply given by the Boltzmann distribution law:

$$g(r) = \exp(-\beta u(r)) \tag{17.10}$$

17.2 Thermodynamic quantities

The total energy is related to the partition function

$$\begin{aligned} E &= -\frac{\partial}{\partial\beta}\ln Q \\ &= -\frac{\partial(\ln Z_N - 3N\ln\lambda - \ln N!)}{\partial\beta} \\ &= \frac{3N}{\lambda}\frac{\partial\lambda}{\partial\beta} - \frac{1}{Z_N}\frac{\partial Z_N}{\partial\beta} \\ &= \frac{3}{2}NkT + \frac{1}{Z_N}\int U(r_1,\cdots,r_N)e^{-\beta U(r_1,\cdots,r_N))}dr_1\cdots dr_N \\ &= \frac{3}{2}NkT + \langle U\rangle \end{aligned} \tag{17.11}$$

where $\lambda = (\beta h^2/2\pi m)^{1/2}$. Therefore,

In order to compute the average energy, one needs to compute the average potential U. Assume the case of a pairwise potential, so that:

$$U(r_1,\cdots,r_N) = \frac{1}{2}\sum_{i\le j} u(|r_i - r_j|) \tag{17.12}$$

$$\begin{aligned} \langle U\rangle &= \frac{1}{2Z_N}\sum_{i\le j}\int u(r_i - r_j)e^{-\beta U(r_1,\cdots,r_N)}dr_1\cdots dr_N \\ &= \frac{N(N-1)}{2Z_N}\int u(r_i - r_j)e^{-\beta U(r_1,\cdots,r_N)}dr_1\cdots dr_N \\ &= \frac{1}{2}\int u(r_1 - r_2)dr_1dr_2\left[\frac{N(N-1)}{Z_N}e^{-\beta U(r_1,\cdots,r_N)}\int dr_3\cdots dr_N\right] \\ &= \frac{1}{2}\int u(r_1 - r_2)dr_1dr_2\rho^{(2)}(r_1,r_2) \\ &= \frac{N^2}{2V^2}\int u(r_1 - r_2)g(r_1,r_2)dr_1dr_2 \\ &= \frac{N^2}{2V}\int 4\pi r^2 u(r)g(r)dr \end{aligned} \tag{17.13}$$

$$E = \frac{3}{2}NkT + \frac{N\rho}{2}\int 4\pi r^2 u(r)g(r)dr \tag{17.14}$$

$$P = -\frac{\partial A}{\partial V} = \frac{1}{\beta}\frac{\partial\ln Q}{\partial V} = \frac{1}{\beta}\frac{\partial(\ln Z_N - 3N\ln\lambda - \ln N!)}{\partial V} = \frac{1}{\beta Z_N}\frac{\partial Z_N}{\partial V} \tag{17.15}$$

We define $s_i = r_i/V^{1/3}$

$$Z_N = V^N\int ds_1\cdots ds_N e^{-\beta U(s_1V^{1/3},\cdots)} \tag{17.16}$$

$$\begin{aligned}\frac{\partial Z_N}{\partial V} &= \frac{N}{V}Z_N - \beta V^N \int ds_1 \cdots ds_N \frac{1}{3V}\sum_{i=1}^{N} r_i \frac{\partial U}{\partial r_i} e^{-\beta U(s_1 V^{1/3},\cdots,-\beta U(s_N V^{1/3})} \\ &= \frac{N}{V}Z_N - \beta \int dr_1 \cdots dr_N \frac{1}{3V}\sum_{i=1}^{N} r_i \cdot F_i e^{-\beta U(r_1,\cdots,r_N)}\end{aligned} \tag{17.17}$$

$$\frac{P}{kT} = \frac{1}{Z_N}\frac{\partial Z_N}{\partial V} = \frac{N}{V} + \frac{\beta}{3V}\langle \sum_{i=1}^{N} r_i \cdot F_i \rangle \tag{17.18}$$

$$\begin{aligned}\frac{\beta}{3V}\langle \sum_{i=1}^{N} r_i \cdot F_i \rangle &= \frac{\beta}{6V}\langle \sum_{i=1}^{N} r_{ij} \cdot F_{ij} \rangle \\ &= \frac{\beta}{6VZ_N}\int \sum_{i\le j} r_{ij} F_{ij} e^{-\beta U(r_1,\cdots,r_N)} \\ &= \frac{\beta N(N-1)}{6VZ_N}\int dr_1 \cdots dr_N r_{12} \cdot F_{12} e^{-\beta U(r_1,\cdots,r_N)} \\ &= \frac{\beta}{6V}\int dr_1 dr_2 r_{12} \cdot F_{12}\left[\frac{N(N-1)}{Z_N}\int dr_3 \cdots dr_N e^{-\beta U(r_1,\cdots,r_N)}\right] \\ &= \frac{\beta}{6V}\int dr_1 dr_2 r_{12} \cdot F_{12}\rho^{(2)}(r_1,r_2) \\ &= \frac{\beta N^2}{6V^3}\int dr_1 dr_2 r_{12} \cdot F_{12} g^{(2)}(r_1,r_2) \\ &= \frac{\beta N^2}{6V^2}\int 4\pi r^3 u'(r) g(r) dr\end{aligned} \tag{17.19}$$

$$\frac{P}{kT} = \rho - \frac{\rho^2}{6kT}\int 4\pi r^3 u'(r) g(r) dr \tag{17.20}$$

This is a general expression, especially useful to evaluate the energy of liquid and fluid.

17.3 Structure factor

The pair correlation function $g(r)$ can be measured experimentally using quasielastic scattering. If a sample is illuminated with a monochromatic beam of x-rays, neutrons, visible light, and so on, the scattered intensity as a function of the angle from the incident and its structure factor. The mathematical definition of the structure factor is:

$$S(\boldsymbol{k}) = \frac{1}{N}\langle \rho_{\boldsymbol{k}} \rho_{-\boldsymbol{k}} \rangle \tag{17.21}$$

where ρ_k is the Fourier transform of $\rho(r)$,

$$\begin{aligned} S(\boldsymbol{k}) &= \frac{1}{N}\langle \iint \exp(-i\boldsymbol{k}(r-r'))\sum_{i=1}^{N}\sum_{j=1,i\neq j}^{N}\delta(r-r_i)\delta(r'-r_j)drdr'\rangle \\ &= 1+\frac{1}{N}\iint \exp(-i\boldsymbol{k}(r-r'))\rho(r,r')drdr' \\ &= 1+\frac{\rho^2}{N}\iint \exp(-i\boldsymbol{k}(r-r'))g(r,r')drdr' \\ &= 1+\rho\int \exp(-i\boldsymbol{k}r)g(r)dr \end{aligned} \tag{17.22}$$

For isotropic fluid, we choose the scalar form,

$$\begin{aligned} S(k) &= 1+2\pi\rho\int r^2 g(r)\int_0^{\pi}\exp(-ikr\cos(\theta))\sin(\theta)d\theta dr \\ &= 1+\rho\int 4\pi r^2 g(r)\frac{\sin(kr)}{kr}dr \end{aligned} \tag{17.23}$$

This states that $S(k)$ is achieved via a Fourier transform of the pair distribution function $g(r)$. Therefore, one can use this formula to run the simulation and compare it with the experimental neutron or X-ray diffraction results.

17.4 Homework

10.4, 10.19